

1h 10m left



ALL



1

2

3

4

4. Special Number

Adarsh had a very special number which he forgot. In past, he had told about this number to one of his friends Abhirav. He went to Abhirav's house and asked about the number. Abhirav is a MindTickler and good at maths so instead of telling about the number directly, he told Adarsh that the number can be obtained by the **XOR** of two numbers between range $[L,R]$. However, this obtained number will be the greatest among the XORs of all pairs between the above-given range. In short, the special number of Adarsh will be the largest possible XORs of all pairs between range $[L,R]$.

Function Description

Complete the `specialNumber` function in the editor below.

`specialNumber` has two parameters:

- L and R which denotes the range as mentioned in the question.

Return

The function must return an integer which denotes the special number of Adarsh.

Constraints

- $1 \leq L \leq R \leq 10^9$

▼ Input Format For Custom Testing

Locked stub code in the editor reads the following input from stdin and passes it to the function:

The first line contains one integer L .

The second line contains one integer R .

▼ Sample Case 0

Sample Input 0

```
1 2
```

Sample Output 0

```
3
```

Explanation 0

All the possible pairs which can be made in range $[1,2]$ are: $\{1,1\}$, $\{1,2\}$, $\{2,2\}$.

Their corresponding XORs are: $\{1^11=0\}$, $\{1^12=3\}$, $\{2^22=0\}$.

▼ Sample Case 1

Sample Input 1

```
2 4
```

Sample Output 1

```
7
```

Explanation 1

All the possible pairs which can be made in range $[2,4]$ are: $\{2,2\}$, $\{2,3\}$, $\{2,4\}$, $\{3,3\}$, $\{3,4\}$, $\{4,4\}$.

Their corresponding XORs are: $\{2^22=0\}$, $\{2^23=1\}$, $\{2^24=6\}$, $\{3^33=0\}$, $\{3^34=7\}$, $\{4^44=0\}$

C++20

Language



Autocomplete Ready

```
1  #include <bits/stdc++.h>
2
3  using namespace std;
4
5  string ltrim(const string &);
6  string rtrim(const string &);
7
8
9
10 /*
11  * Complete the 'specialNumber' function below.
12  *
13  * The function is expected to return an INTEGER.
14  * The function accepts following parameters:
15  * 1. INTEGER l
16  * 2. INTEGER r
17  */
18
19 int specialNumber(int l, int r) {
20
21 }
22
23 > int main() ...
```

Line: 1 Col: 25